



Chapter 22 – Project Management

Topics covered



- ✧ Risk management
- ✧ Managing people
- ✧ Teamwork

Software project management



- ✧ Concerned with activities involved in ensuring that software is delivered on time and on schedule and in accordance with the requirements of the organisations developing and procuring the software.
- ✧ Project management is needed because software development is always subject to budget and schedule constraints that are set by the organisation developing the software.

Success criteria



- ✧ Deliver the software to the customer at the agreed time.
- ✧ Keep overall costs within budget.
- ✧ Deliver software that meets the customer's expectations.
- ✧ Maintain a coherent and well-functioning development team.



Risk management

Risk management



- ✧ Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project.
- ✧ Software risk management is important because of the inherent uncertainties in software development.
 - These uncertainties stem from loosely defined requirements, requirements changes due to changes in customer needs, difficulties in estimating the time and resources required for software development, and differences in individual skills.
- ✧ You have to anticipate risks, understand the impact of these risks on the project, the product and the business, and take steps to avoid these risks.

Risk classification



- ✧ There are two dimensions of risk classification
 - The type of risk (technical, organizational, ..)
 - what is affected by the risk:
- ✧ *Project risks* affect schedule or resources;
- ✧ *Product risks* affect the quality or performance of the software being developed;
- ✧ *Business risks* affect the organisation developing or procuring the software.

Examples of project, product, and business risks



Risk	Affects	Description
Staff turnover	Project	Experienced staff will leave the project before it is finished.
Management change	Project	There will be a change of organizational management with different priorities.
Hardware unavailability	Project	Hardware that is essential for the project will not be delivered on schedule.
Requirements change	Project and product	There will be a larger number of changes to the requirements than anticipated.
Specification delays	Project and product	Specifications of essential interfaces are not available on schedule.
Size underestimate	Project and product	The size of the system has been underestimated.
CASE tool underperformance	Product	CASE tools, which support the project, do not perform as anticipated.
Technology change	Business	The underlying technology on which the system is built is superseded by new technology.
Product competition	Business	A competitive product is marketed before the system is completed.

The risk management process



✧ Risk identification

- Identify project, product and business risks;

✧ Risk analysis

- Assess the likelihood and consequences of these risks;

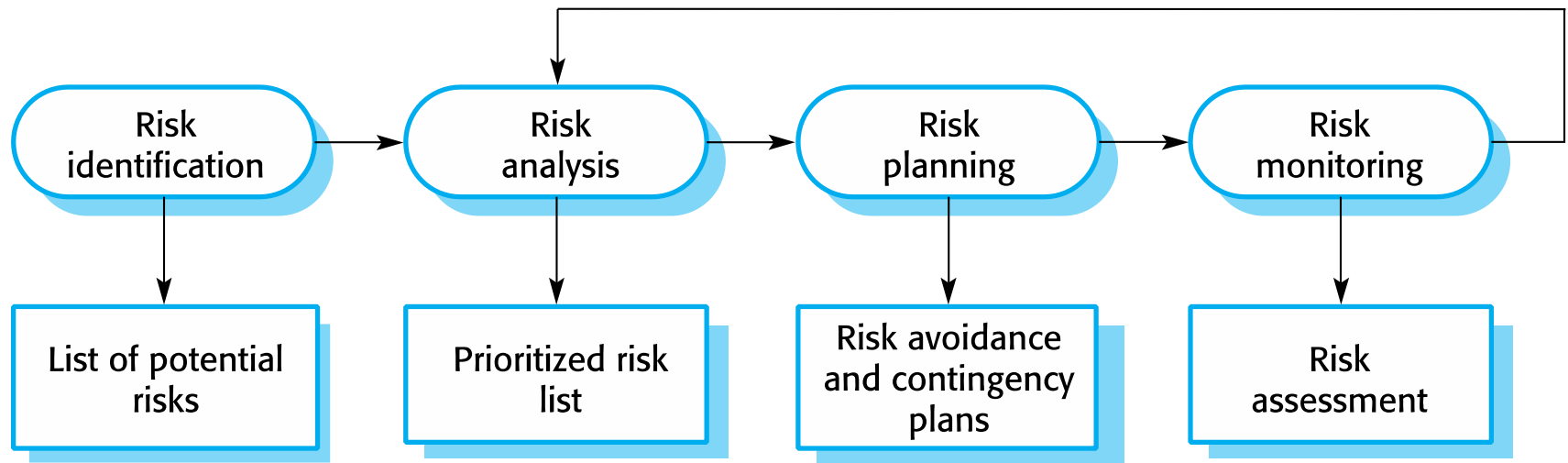
✧ Risk planning

- Draw up plans to avoid or minimise the effects of the risk;

✧ Risk monitoring

- Monitor the risks throughout the project;

The risk management process



Risk identification



- ✧ May be a team activities or based on the individual project manager's experience.
- ✧ A checklist of common risks may be used to identify risks in a project
 - Technology risks.
 - Organizational risks.
 - People risks.
 - Requirements risks.
 - Estimation risks.

Examples of different risk types



Risk type	Possible risks
Estimation	The time required to develop the software is underestimated. (12) The rate of defect repair is underestimated. (13) The size of the software is underestimated. (14)
Organizational	The organization is restructured so that different management are responsible for the project. (6) Organizational financial problems force reductions in the project budget. (7)
People	It is impossible to recruit staff with the skills required. (3) Key staff are ill and unavailable at critical times. (4) Required training for staff is not available. (5)
Requirements	Changes to requirements that require major design rework are proposed. (10) Customers fail to understand the impact of requirements changes. (11)
Technology	The database used in the system cannot process as many transactions per second as expected. (1) Reusable software components contain defects that mean they cannot be reused as planned. (2)
Tools	The code generated by software code generation tools is inefficient. (8) Software tools cannot work together in an integrated way. (9)

Risk analysis



- ✧ Assess probability and seriousness of each risk.
- ✧ Probability may be very low, low, moderate, high or very high.
- ✧ Risk consequences might be catastrophic, serious, tolerable or insignificant.

Risk types and examples



Risk	Probability	Effects
Organizational financial problems force reductions in the project budget (7).	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project (3).	High	Catastrophic
Key staff are ill at critical times in the project (4).	Moderate	Serious
Faults in reusable software components have to be repaired before these components are reused. (2).	Moderate	Serious
Changes to requirements that require major design rework are proposed (10).	Moderate	Serious
The organization is restructured so that different management are responsible for the project (6).	High	Serious
The database used in the system cannot process as many transactions per second as expected (1).	Moderate	Serious

Risk planning



- ✧ Consider each risk and develop a strategy to manage that risk.
- ✧ Avoidance strategies
 - The probability that the risk will arise is reduced;
- ✧ Minimization strategies
 - The impact of the risk on the project or product will be reduced;
- ✧ Contingency plans
 - If the risk arises, contingency plans are plans to deal with that risk;

What-if questions



- ✧ What if several engineers are ill at the same time?
- ✧ What if an economic downturn leads to budget cuts of 20% for the project?
- ✧ What if the performance of open-source software is inadequate and the only expert on that open source software leaves?
- ✧ What if the company that supplies and maintains software components goes out of business?
- ✧ What if the customer fails to deliver the revised requirements as predicted?

Strategies to help manage risk



Risk	Strategy
Organizational financial problems	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business and presenting reasons why cuts to the project budget would not be cost-effective.
Recruitment problems	Alert customer to potential difficulties and the possibility of delays; investigate buying-in components.
Staff illness	Reorganize team so that there is more overlap of work and people therefore understand each other's jobs.
Defective components	Replace potentially defective components with bought-in components of known reliability.
Requirements changes	Derive traceability information to assess requirements change impact; maximize information hiding in the design.

Strategies to help manage risk



Risk	Strategy
Organizational restructuring	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Database performance	Investigate the possibility of buying a higher-performance database.
Underestimated development time	Investigate buying-in components; investigate use of a program generator.

Risk monitoring



- ✧ Assess each identified risks regularly to decide whether or not it is becoming less or more probable.
- ✧ Also assess whether the effects of the risk have changed.
- ✧ Each key risk should be discussed at management progress meetings.

Risk indicators



Risk type	Potential indicators
Estimation	Failure to meet agreed schedule; failure to clear reported defects.
Organizational	Organizational gossip; lack of action by senior management.
People	Poor staff morale; poor relationships amongst team members; high staff turnover.
Requirements	Many requirements change requests; customer complaints.
Technology	Late delivery of hardware or support software; many reported technology problems.
Tools	Reluctance by team members to use tools; complaints about CASE tools; demands for higher-powered workstations.



Managing people

Managing people



- ✧ People are an organisation's most important assets.
- ✧ The tasks of a manager are essentially people-oriented. Unless there is some understanding of people, management will be unsuccessful.
- ✧ Poor people management is an important contributor to project failure.

People management factors



✧ Consistency

- Team members should all be treated in a comparable way without favourites or discrimination.

✧ Respect

- Different team members have different skills and these differences should be respected.

✧ Inclusion

- Involve all team members and make sure that people's views are considered.

✧ Honesty

- You should always be honest about what is going well and what is going badly in a project.

Motivating people



- ✧ An important role of a manager is to motivate the people working on a project.
- ✧ Motivation means organizing the work and the working environment to encourage people to work effectively.
 - If people are not motivated, they will not be interested in the work they are doing. They will work slowly, be more likely to make mistakes and will not contribute to the broader goals of the team or the organization.
- ✧ Motivation is a complex issue but it appears that there are different types of motivation based on:
 - Basic needs (e.g. food, sleep, etc.);
 - Personal needs (e.g. respect, self-esteem);
 - Social needs (e.g. to be accepted as part of a group).

Need satisfaction



- ✧ In software development groups, basic physiological and safety needs are not an issue.
- ✧ Social
 - Provide communal facilities;
 - Allow informal communications e.g. via social networking
- ✧ Esteem
 - Recognition of achievements;
 - Appropriate rewards.
- ✧ Self-realization
 - Training - people want to learn more;
 - Responsibility.

Personality types



- ✧ The needs hierarchy is almost certainly an oversimplification of motivation in practice.
- ✧ Motivation should also take into account different personality types:
 - Task-oriented people, who are motivated by the work they do. In software engineering.
 - Interaction-oriented people, who are motivated by the presence and actions of co-workers.
 - Self-oriented people, who are principally motivated by personal success and recognition.

Personality types



✧ Task-oriented.

- The motivation for doing the work is the work itself;

✧ Self-oriented.

- The work is a means to an end which is the achievement of individual goals - e.g. to get rich, to play tennis, to travel etc.;

✧ Interaction-oriented

- The principal motivation is the presence and actions of co-workers. People go to work because they like to go to work.

Motivation balance



- ✧ Individual motivations are made up of elements of each class.
- ✧ The balance can change depending on personal circumstances and external events.
- ✧ However, people are not just motivated by personal factors but also by being part of a group and culture.
- ✧ People go to work because they are motivated by the people that they work with.



Teamwork

Teamwork



- ✧ Most software engineering is a group activity
 - The development schedule for most non-trivial software projects is such that they cannot be completed by one person working alone.
- ✧ A good group is cohesive and has a team spirit. The people involved are motivated by the success of the group as well as by their own personal goals.
- ✧ Group interaction is a key determinant of group performance.
- ✧ Flexibility in group composition is limited
 - Managers must do the best they can with available people.

Group cohesiveness



- ✧ In a cohesive group, members consider the group to be more important than any individual in it.
- ✧ The advantages of a cohesive group are:
 - Group quality standards can be developed by the group members.
 - Team members learn from each other and get to know each other's work; Inhibitions caused by ignorance are reduced.
 - Knowledge is shared. Continuity can be maintained if a group member leaves.
 - Refactoring and continual improvement is encouraged. Group members work collectively to deliver high quality results and fix problems, irrespective of the individuals who originally created the design or program.

The effectiveness of a team



✧ The people in the group

- You need a mix of people in a project group as software development involves diverse activities such as negotiating with clients, programming, testing and documentation.

✧ The group organization

- A group should be organized so that individuals can contribute to the best of their abilities and tasks can be completed as expected.

✧ Technical and managerial communications

- Good communications between group members, and between the software engineering team and other project stakeholders, is essential.

Selecting group members



- ✧ A manager or team leader's job is to create a cohesive group and organize their group so that they can work together effectively.
- ✧ This involves creating a group with the right balance of technical skills and personalities, and organizing that group so that the members work together effectively.

Assembling a team



- ✧ May not be possible to appoint the ideal people to work on a project
 - Project budget may not allow for the use of highly-paid staff;
 - Staff with the appropriate experience may not be available;
 - An organisation may wish to develop employee skills on a software project.
- ✧ Managers have to work within these constraints especially when there are shortages of trained staff.

Group composition



- ✧ Group composed of members who share the same motivation can be problematic
 - Task-oriented - everyone wants to do their own thing;
 - Self-oriented - everyone wants to be the boss;
 - Interaction-oriented - too much chatting, not enough work.
- ✧ An effective group has a balance of all types.
- ✧ This can be difficult to achieve software engineers are often task-oriented.
- ✧ Interaction-oriented people are very important as they can detect and defuse tensions that arise.